AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII EXAMINATION

APRIL/ MAY 2019

Business Statistics Paper II

Time: 1 hour 30 minutes Marks: 30

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INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's Signature**

RUBRIC

2. There are SEVEN questions. Answer ALL questions. Choices are specified inside the paper.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a scientific calculator if you wish.

Q.1. (Total 4 Marks)

i. Write any TWO points to highlight the importance of statistics in different fields. (2 Marks)

ii. Write any TWO limitations of statistics. (2 Marks)

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Q.2. (Total 4 Marks)

In an under 19 cricket camp, the weights of 25 players were recorded as follows:

35, 40, 40, 40, 45, 45, 45, 44, 37, 37, 36, 50, 50, 38, 38, 39, 41, 46, 47, 47, 48, 43, 43, 42, 42

i. Use the given table to construct the frequency distribution of the given data. (2 Marks)

ii. Also complete the column for cumulative frequency and class mark. (2 Marks)

Weight (kg)	Number of Participants	Cumulative Frequency	Class Mark
35 – 37			
38 – 40			
41 – 43			
44 – 46			
47 – 49			
50 – 52			

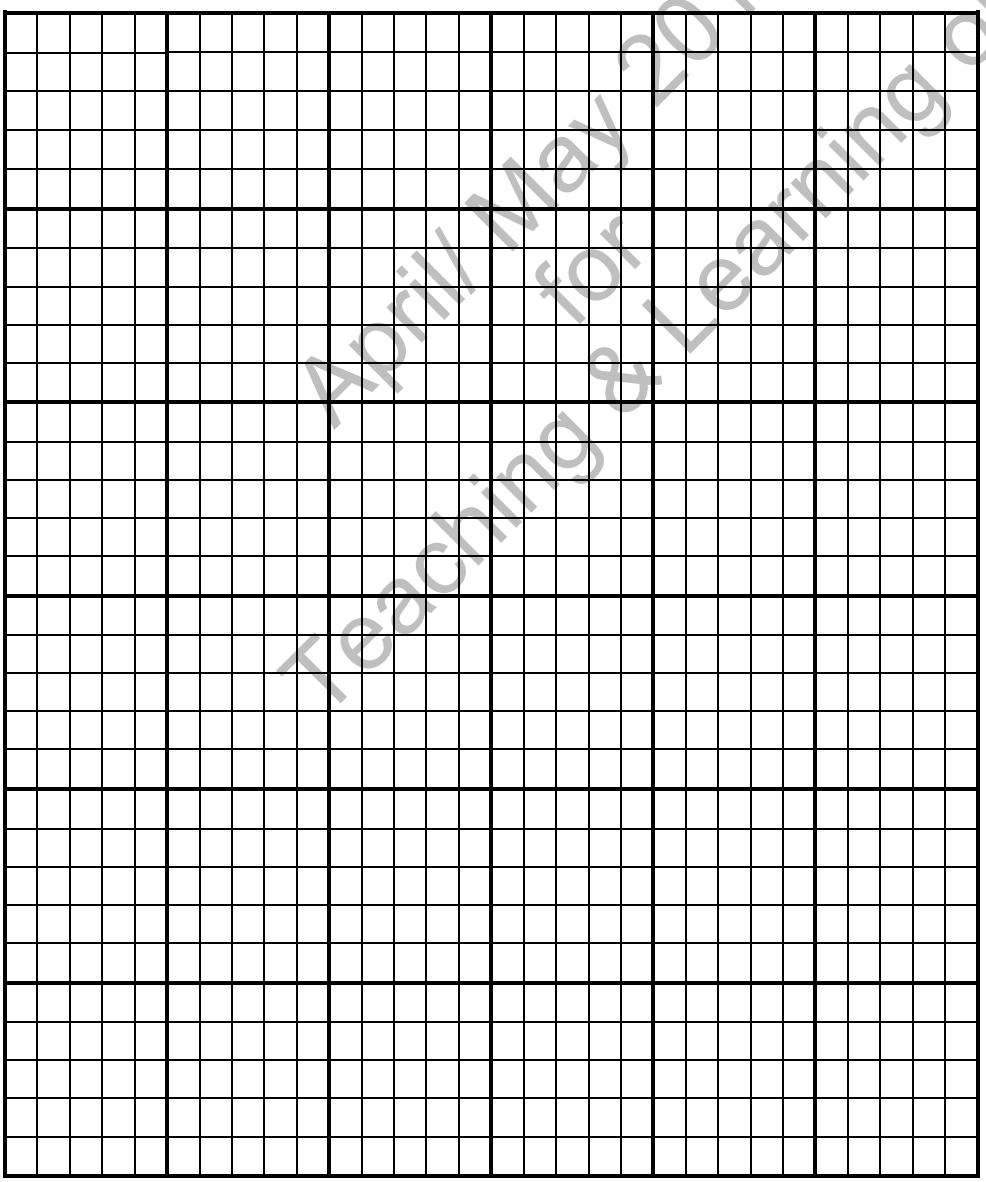
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Q.3. (Total 4 Marks)

The given table shows the yields, in thousand kilograms, of wheat, rice and gram. These commodities are produced at a certain agricultural farm from years 2010 to 2013.

Year	Wheat (kg)	Rice (kg)	Gram (kg)
2010	65	35	20
2011	65	40	25
2012	60	40	25
2013	70	30	20

On the given graph, construct a component/ subdivided bar chart to represent the data.



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(ATTEMPT EITHER PART a OR PART b OF Q.4.)

Q.4. (Total 5 Marks)

a. In the given table, Bilal recorded the time in seconds of 30 people to complete a race. (5 Marks)

Time (s)	**Number of People**	
51 – 55	4	
56 – 60	10	
61 – 65	11	
66 – 70	5	

Label and fill the empty column to find the median of this data.

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(**ATTEMPT EITHER PART a OR PART b OF Q.4.)**

b. The given data shows the ages of 30 people working in a department. Label and fill empty column of the table to calculate the mode of the data and interpret your answer. (5 Marks)

Age (year)	**Number of People**	
31 – 40	9	
41 – 50	5	
51 – 60	10	
61 – 70	6	

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Q.5. (Total 4 Marks)

In a nursery, the lengths of 80 plants of certain species were measured in centimetres. These measurements were recorded in the given table.

Length of Plant (cm)	**Number of Plants**	**Class Mark**		
15 – 19	12	17		
20 – 24	17	22		
25 – 29	28	27		
30 – 34	13	32		
35 – 39	10	37		
Total	**80**	-	-	

For the given data, the mean height is 26.5 cm. Fill the empty column of the table to calculate mean deviation of the heights of the plants.

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(ATTEMPT EITHER PART a OR PART b OF Q.6.)

Q.6. (Total 5 Marks)

a.

i. Write any TWO steps of construction of price index. (1 Mark)

ii. Describe any TWO steps of construction of price index. (4 Marks)

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(ATTEMPT EITHER PART a OR PART b OF Q.6.)

b. For the given data in the table, calculate Paasche's index number using base year 2010. Also interpret your result. (5 Marks)

	2010		2015			
Commodity	**Quantity (kg)**	**Price/ kg (Rs)**	**Quantity (kg)**	**Price/ kg (Rs)**		
Wheat	100	10	120	13		
Rice	50	100	40	110		
Cotton	35	148	40	160		
Butter	12	600	10	750		
Mutton	48	500	35	600		
Total	-	-	-	-		

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(ATTEMPT EITHER PART a OR PART b OF Q.7.)

Q.7. (Total 4 Marks)

a.

i. How many passwords of four letters are possible by using the letters of the word 'SOUTH', if repetition is

I. allowed? (1 Mark)

II. NOT allowed? (1 Mark)

ii. There were 9 persons in the waiting list to book a flight to Gilgit. If 5 seats are cancelled and are now available, then find the possible number of ways of selection of 5 people from the waiting list. (2 Marks)

b. A bag contains eleven cards. Each letter of the word MATHEMATICS is written on a separate card kept in the bag. If two cards are randomly drawn without replacement, then find the probability that

i. both cards have letter M. (2 Marks)

ii the first card has letter M or second card has letter T. (2 Marks)

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END OF PAPER

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